

Contribution Statement

Team: Software_Final-Project_M201 (Dilara Rzazada / Melek Askerova / Sanan Hajiyev / Nezrin Ibrahimova)

Course: AI-ENG-110 — Software Engineering — Autumn 2026

Topic: Topic 4 — Async Research Assistant

Repository: https://github.com/Dilara898/Software_Final-Project_M201

Final tag: v1.0-final

Submission date: 2026-09-19

Commit and PR ownership summary

This statement follows the repository's pull request history (32 closed pull requests, 0 open, at the time of writing). Each pull request is assigned to the member who authored it. Commit shares come from `git shortlog -sn --no-merges` on main at the time of writing (70 non-merge commits).

PR	Title / work item	Main owner
#4	feat: add storage layer (postgres, cache, history)	Dilara Rzazada
#5	fix: harden C cancellation and benchmark validation	Sanan Hajiyev
#6	fix: harden C cancellation and benchmark validation (second PR with the same title)	Sanan Hajiyev
#7	test: add cache store and config tests	Dilara Rzazada
#8	Feat/d.cli — CLI commands	Nezrin Ibrahimova
#9	Feat/b ai service — AI service	Melek Askerova
#10	fix(b): address remaining review findings B-05, B-06, B-07 + logging	Melek Askerova
#11	fix: lazy settings + concurrency safety in storage layer	Dilara Rzazada
#12	feat(c): integrate A storage and B services with research workflow	Sanan Hajiyev
#13	test(c): stabilize cancellation and cleanup tests	Sanan Hajiyev
#14	Feat/d cli entrypoint — CLI entry point	Nezrin Ibrahimova
#15	fix: follow arXiv redirect from one shared source client	Sanan Hajiyev
#16	Problem/timeout 10	Nezrin Ibrahimova
#17	feat(ui): add Streamlit UI reusing the CLI research pipeline	Melek Askerova
#18	Fix arXiv HTTPS requests and add HTTP regression coverage	Sanan Hajiyev
#19	UI service	Melek Askerova
#20	docs(report): draft final project report from REPORT_TEMPLATE	Melek Askerova
#21	UI service	Melek Askerova
#22	Update ui.py	Dilara Rzazada
#23	Update application.py	Dilara Rzazada
#24	fix: add google-genai package required by Gemini provider	Dilara Rzazada
#25	Update orchestrator.py	Dilara Rzazada
#26	Update ui.py	Dilara Rzazada
#27	fix: repair broken test suite and restore pinned dependencies	Sanan Hajiyev
#28	Update README.md	Dilara Rzazada
#29	fix(b): flag suspicious empty Wikipedia results distinctly in logs	Melek Askerova
#30	Feat/d validation tests	Nezrin Ibrahimova
#31	fix(ui): stop showing internal cache-layer warnings as user-facing failures	Melek Askerova
#32	Fix Streamlit deployment reloads and remote cache deadlines	Sanan Hajiyev
#[]	[remaining pull requests not yet listed]	[]

Member A — Dilara Rzazada (@Dilara898)

Owned (sole author of these files / PRs):

- researcher/storage/ — asyncpg connection pool, CacheStore ABC with in-memory and PostgreSQL implementations, session history, schema.sql (PR #4)
- researcher/config.py — lazy typed settings and storage concurrency safety (PR #11)
- Cache store and config tests (PR #7)
- Final report preparation (docs/)
- PRs authored: #4, #7, #11, #22, #23, #24, #25, #26, #28

Co-owned (paired or substantially edited):

- researcher/ui.py (main author: Member B) — later updates in PRs #22 and #26
- researcher/application.py (main author: Member D) — update in PR #23
- researcher/concurrency/orchestrator.py (main author: Member C) — update in PR #25
- README.md (main author: Member B) — update in PR #28
- Requirements files — added the google-genai package required by the Gemini provider (PR #24), with Member C (PR #27)
- Storage adapters that connect the workflow layer to the storage layer (with Member C)
- PostgreSQL service in docker-compose.yml and schema initialization (with Member D)

Reviewed (PRs reviewed and merged):

- PRs: #[]

Approximate share of commits: 27% (19 of 70 non-merge commits; two git identities: Dilara898 12 and Dilara Rzazada 7)

Member B — Melek Askerova (@askerovamelek)

Owned (sole author of these files / PRs):

- researcher/services/ai_service.py — AIFetchService and AISynthesisService: the only place the code crosses the provided ai/ boundary (PR #9)
- Retry and pacing mechanisms, structured logging, and the follow-up review fixes B-05, B-06, B-07 (PR #10)
- Distinct log flag for suspicious empty Wikipedia results (PR #29)
- UI fix: internal cache-layer warnings no longer shown as user-facing failures (PR #31)
- Service tests
- PRs authored: #9, #10, #17, #19, #20, #21, #29, #31

Co-owned (paired or substantially edited):

- researcher/ui.py — Streamlit UI (Ask and History views): main author (PRs #17, #19, #21, #31); later updates by Member A (PRs #22, #26)
- README.md — documentation: main author; final update by Member A (PR #28)
- Initial draft of the final report from REPORT_TEMPLATE (PR #20); the final report was prepared by Member A
- Fetch-service HTTP behavior after the shared HTTP client was introduced (with Member C)

Reviewed (PRs reviewed and merged):

- PRs: #[]

Approximate share of commits: 26% (18 of 70 non-merge commits)

Member C — Sanan Hajiyeu (@20hajiyeu)

Owned (sole author of these files / PRs):

- researcher/concurrency/ — SourceOrchestrator (bounded concurrent collection, per-source timeouts, cache and timing outcomes) and ResearchOrchestrator (collect, synthesize, validate, persist); integration of Member A's storage and Member B's services into the research workflow (PR #12)

- Cancellation hardening and benchmark validation (PRs #5, #6), and the cancellation and cleanup tests (PR #13)
- Shared HTTP client, arXiv redirect and HTTPS fixes, and HTTP regression tests (PRs #15, #18)
- Test-suite repair and restoration of pinned dependencies (PR #27)
- Streamlit deployment reload and remote cache deadline fixes (PR #32)
- CONTRIBUTION_STATEMENT.md
- PRs authored: #5, #6, #12, #13, #15, #18, #27, #32

Co-owned (paired or substantially edited):

- researcher/concurrency/orchestrator.py — update by Member A (PR #25)
- Storage integration (with Member A)
- Fetch-service HTTP behavior (with Member B)

Reviewed (PRs reviewed and merged):

- PRs: #[]

Approximate share of commits: 20% (14 of 70 non-merge commits)

Member D — Nezrin Ibrahimova (@nezrinibrahimova)

Owned (sole author of these files / PRs):

- researcher/cli.py — CLI commands ask and history (PR #8)
- CLI entry point, researcher/__main__.py (PR #14)
- Timeout-related change (PR #16)
- researcher/application.py — composition root (build_result) shared by the CLI and the UI — PR #[]
- researcher/validation.py — input validation — PR #[]
- Dockerfile and Docker/Compose integration — PR #[]
- scripts/demo.py — demo scripts — PR #[]
- Input validation tests (tests/test_d_validation.py) (PR #30)
- CLI and integration tests — PR #[]
- Presentation preparation — PR/commit #[]
- PRs authored: #8, #14, #16, #30

Co-owned (paired or substantially edited):

- researcher/application.py — update by Member A (PR #23)
- docker-compose.yml PostgreSQL service and schema initialization (with Member A)
- application.build_result used by the Streamlit UI (with Member B)

Reviewed (PRs reviewed and merged):

- PRs: #[]

Approximate share of commits: 27% (19 of 70 non-merge commits)

AI tool disclosure (also in §9 of the report)

We used AI coding assistants for the following work.

Member	Assistant	How it helped
Dilara Rzazada (A)	Gemini	Helped write configuration and storage code, and assisted with debugging and testing.
Melek Askerova (B)	Claude	Helped write AI-service and UI code, and assisted with debugging and testing.
Sanan Hajiyev (C)	Codex	Helped write concurrency and orchestration code, and assisted with debugging and testing.
Nezrin Ibrahimova (D)	Codex	Helped write CLI and application-integration code, and assisted with debugging and testing.

We affirm that we **can defend every line of code** in this repository during the oral defense. "The AI wrote it" is not an answer we will use.

Signatures

By signing below, we affirm that:

- The contributions described above are accurate.
- The commit percentages reflect actual work, not artificially split commits.
- Every line of code in the repository can be defended by at least one team member.
- AI assistant usage has been disclosed as described above.

Member	Signature	Date
Dilara Rzazada	_____	2026-09-19
Melek Askerova	_____	2026-09-19
Sanan Hajiyev	_____	2026-09-19
Nezrin Ibrahimova	_____	2026-09-19